Karen Xiao

Class of 2026 CS & Math Student at Wellesley College

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EDUCATION

Wellesley College Wellesley, MA
B.A. Computer Science (Honors) and Mathematics, GPA: 3.91/4.0 *May 2026*

Massachusetts Institute of Technology Cambridge, MA
Cross-Registered – 6.390 Introduction to Machine Learning *Fall 2024*

University of Maryland, College Park College Park, MD
Honors College, Computer Science, GPA: 3.97/4.0 *August 2022 – May 2023*
Gemstone Research Program

RELEVANT COURSES

Human Computer Interaction; Natural Language Processing; Fundamental Algorithms; Deep Learning; Introduction to Machine Learning; Regression Analysis & Statistical Models; Theory of Computation; Computer Systems; Data Structures; Combinatorics & Graph Theory; Advanced Linear Algebra; Abstract Algebra; Multivariable Calculus; Real Analysis

RESEARCH EXPERIENCE

Senior Honors Thesis Wellesley College
Advisor: Brian Brubach *August 2025 – present*

- Completed a senior honors thesis that proposes methods to measure the diversity of evaluation in classes by treating different types of assignments as alternative ways of ranking students

NSF REU Intern Carnegie Mellon University
Human Computer Interaction Institute *May 2025 – September 2025*

- Worked with Conrad Borchers and Dr. Vincent Aleven in the Creating Adaptive Tutoring Systems (CATS) Lab
- Applied statistical methods to analyze the impact of data-driven redesign on intelligent tutoring systems
- Identified new insights into what constitutes a productive level of difficulty for data-optimized practice

Research Assistant Wellesley College
Institute for Mathematics and Democracy *December 2024 – September 2025*

- Simulated various ranked choice election methods on datasets containing ballots for thousands of real elections
- Conducted empirical comparison of various ranked choice voting methods using Python and VoteKit
- This research is believed to be the largest study of ranked choice elections ever performed

Undergraduate Student Researcher Wellesley College
CrowdNets Lab *August 2024 - May 2025*

- Defined and evaluated a graph-based spatio-temporal POI ranking model that outperforms baseline ML models
- Ran simulations and designed dynamic visualizations on large-scale real-world Uber data to evaluate performance
- Won 2nd place in the Undergraduate Student Research Competition at ACM SIGSPATIAL 2025 (Minneapolis, MN)

Undergraduate Student Researcher MIT
MIT Urban Metabolism Group *February 2024 - May 2024*

- Collected and analyzed data regarding climate risks and policies in various metropolitan clusters
- Converted a policy portfolio into a quantifiable matrix for ML-driven insights

PUBLICATIONS

Liu, Q., Borchers, C., Xia, M., **Xiao, K.,** Carvalho, P., Koedinger, K., & Aleven, V. (in press). Evaluating a Process for the Data-driven Redesign of Educational Technology: Improved Learning Processes. In *The 27th International Conference on AI in Education (AIED '26).*

Xiao, K., Irisumi, M., & Bassem C. (2025). A Graph-Based Spatio-Temporal POI Ranking Measure for Pickup and Delivery Platforms. In *The 33rd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '25).* https://doi.org/10.1145/3748636.3766535 **[SRC 2nd Place Winner]**

Xiao, K. (2023). "Normalize Menstruation, End Period Poverty" *UMD Interpolations (Fall 2023 ed.*). https://english.umd.edu/research-innovation/journals/interpolations/fall-2023/normalize-menstruation-end-period-poverty

WORK EXPERIENCE

Financial Data Science Internship — Quantbot Technologies LP
Data Science Intern — *May 2024 - August 2024*

- Optimized data capture and validation process using DuckDB, Pandas, and Apache Iceberg
- Assessed, enriched, and normalized company outlook and projection data covering all global markets
- Researched new database query tools that the company can implement to optimize data capture toolkits

Financial Data Science Internship — Quantbot Technologies LP
Data Science Intern — *May 2023 - August 2023*

- Validated and refined raw Equities and Futures data using Python Pandas and NumPy on AWS
- Collaborated with quantitative researchers, traders, and external data vendors
- Optimized existing data processing toolkits using SQLite and Pandas

LEADERSHIP EXPERIENCE

Undergraduate Teaching Assistant — Wellesley College
Wellesley College Computer Science Department — *January 2024 - present*

- Lead weekly office hours, offer 1-on-1 tutoring, and grade assignments for CS 231 Design & Analysis of Algorithms over the course of 4 semesters

Captain — Wellesley College
Wellesley College Club Squash Team — *August 2025 - present*

- Lead and organize a team of 15 players by coordinating practice and matches, mentoring new players, and fostering a positive team environment

Networking Chair — Wellesley College
Wellesley Computer Science Club — *October 2024 - present*

- Manage alumni and recruiting connections to organize and communicate important recruiting, networking, and volunteering events to underrepresented students interested in CS opportunities

AWARDS, FELLOWSHIPS, & GRANTS

Conference Funding

- (2025) **SIGSPATIAL Travel Award (SIGSPATIAL '26) - $1,000,** National Science Foundation
- (2025) **Science Center Student Conference Travel Grant - $300,** Wellesley College